Grating k-space Diagram for RGB FOV

GUI User Guide

Yen-Chi Chung (Angel)

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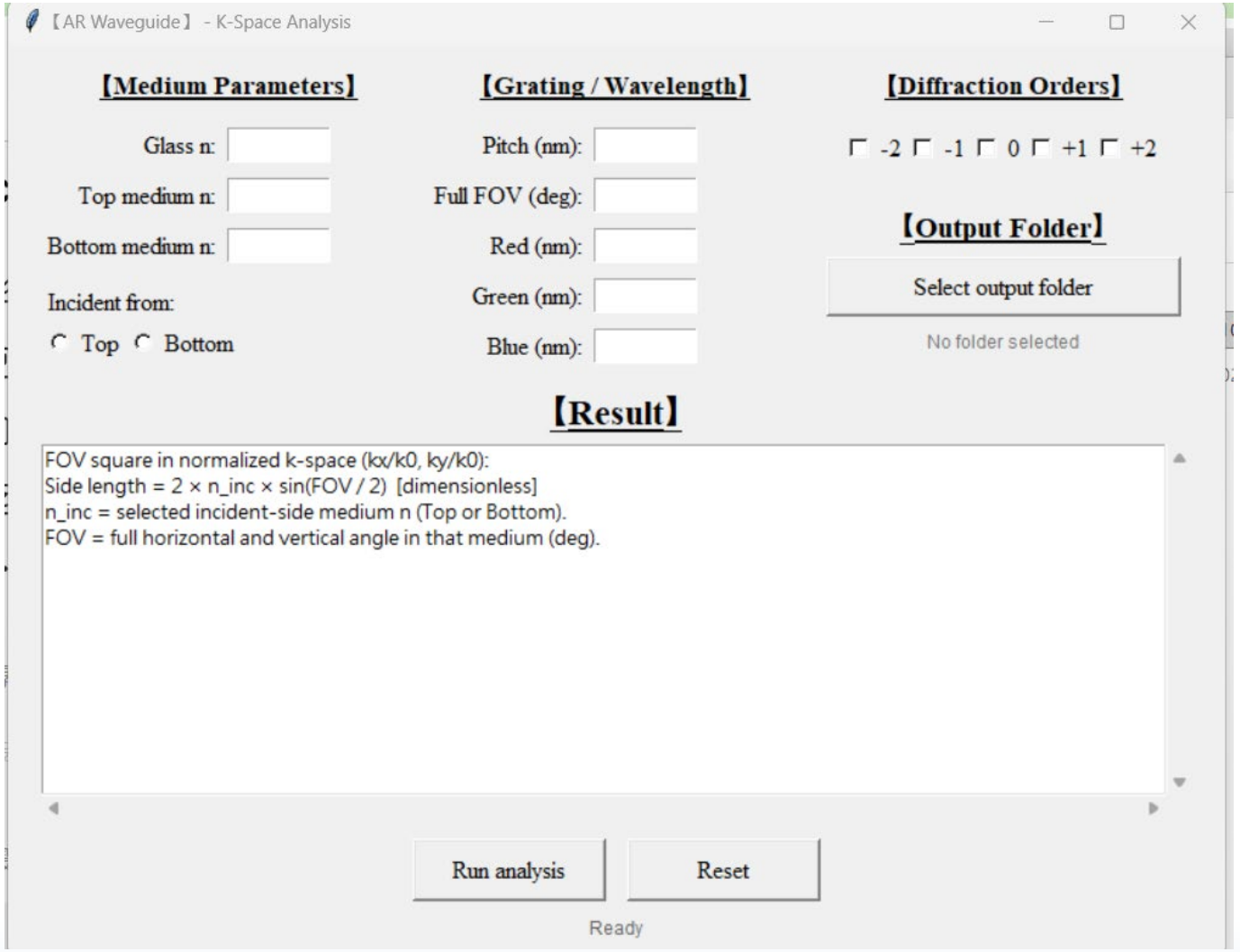
Grating k-space – GUI

Purpose and initial screen

Visualize RGB diffraction orders in normalized k-space.

Check whether the entire FOV square lies within the TIR region.

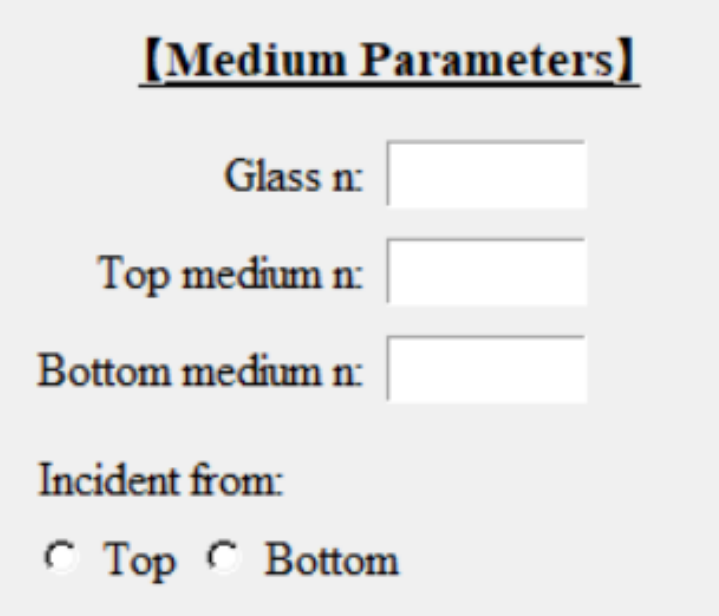
All inputs start blank. Select the incident side and diffraction orders manually.



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1. Medium parameters and incident side



Field	Meaning
Glass n	Refractive index of the waveguide glass
Top medium n	Medium touching the top glass surface
Bottom medium n	Medium touching the bottom glass surface
Incident from	Top or Bottom selects the corresponding n_inc

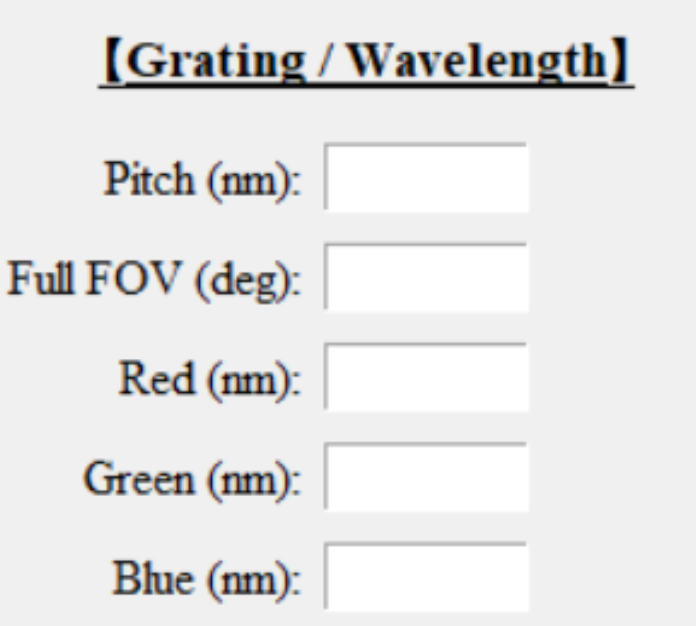
Example: Top = Bottom = 1.0; Glass = 2.2; incident from Bottom.

Glass n must be greater than both surrounding-medium indices.

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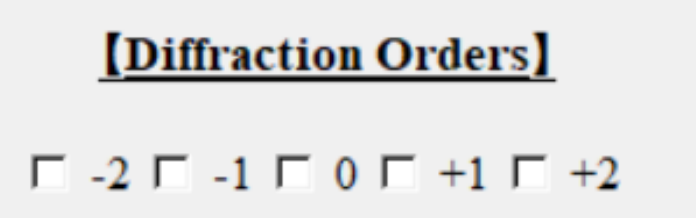
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2. Pitch, RGB wavelengths and diffraction orders



Pitch (nm): the grating period along x.

Red / Green / Blue (nm): vacuum wavelengths.



Select any of −2, −1, 0, +1, +2. At least one order is required.

Example: pitch 350 nm; RGB 630 / 530 / 450 nm; select 0 and +1.

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3. Full FOV and square size

u = kx/k0 v = ky/k0

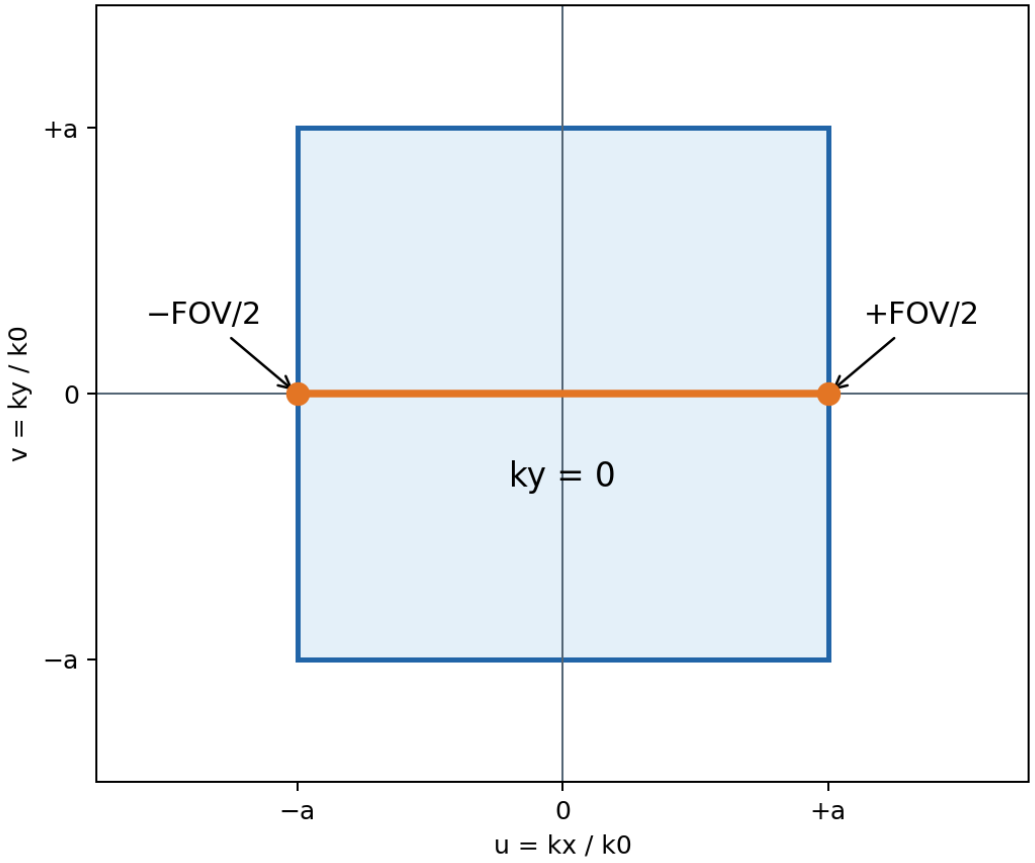
a = n_inc sin(FOV/2)

Square side length = 2a

Both axes span −a to +a.

One FOV input applies to both directions.

Example: FOV = 40°, n_inc = 1
a = 0.34202; side = 0.68404.



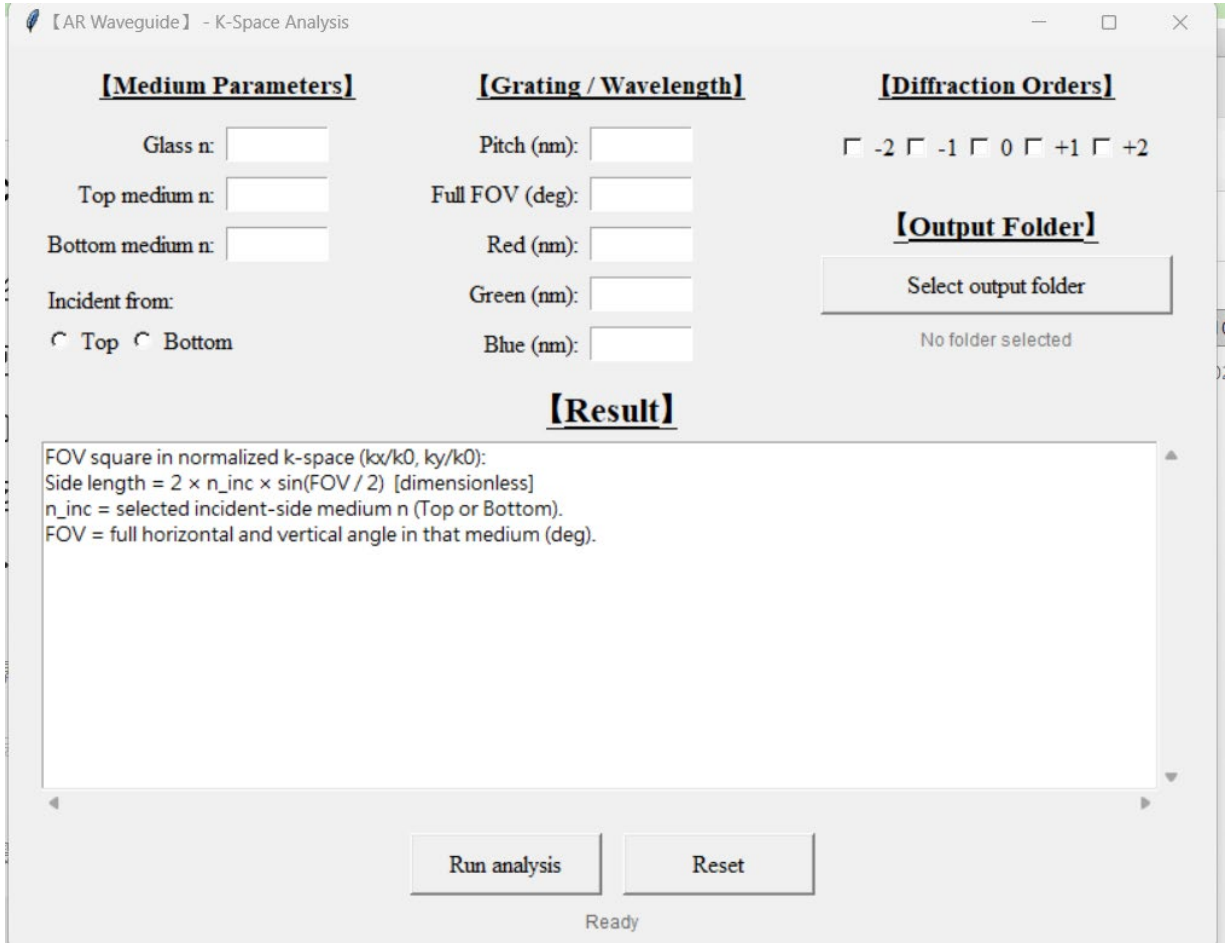
Coordinates are dimensionless. The square describes direction coverage, not the physical beam footprint.

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4. Select output folder and run

1. Enter all numeric parameters.
2. Choose Top or Bottom.
3. Select diffraction orders.
4. Click Select output folder.
5. Click Run analysis.

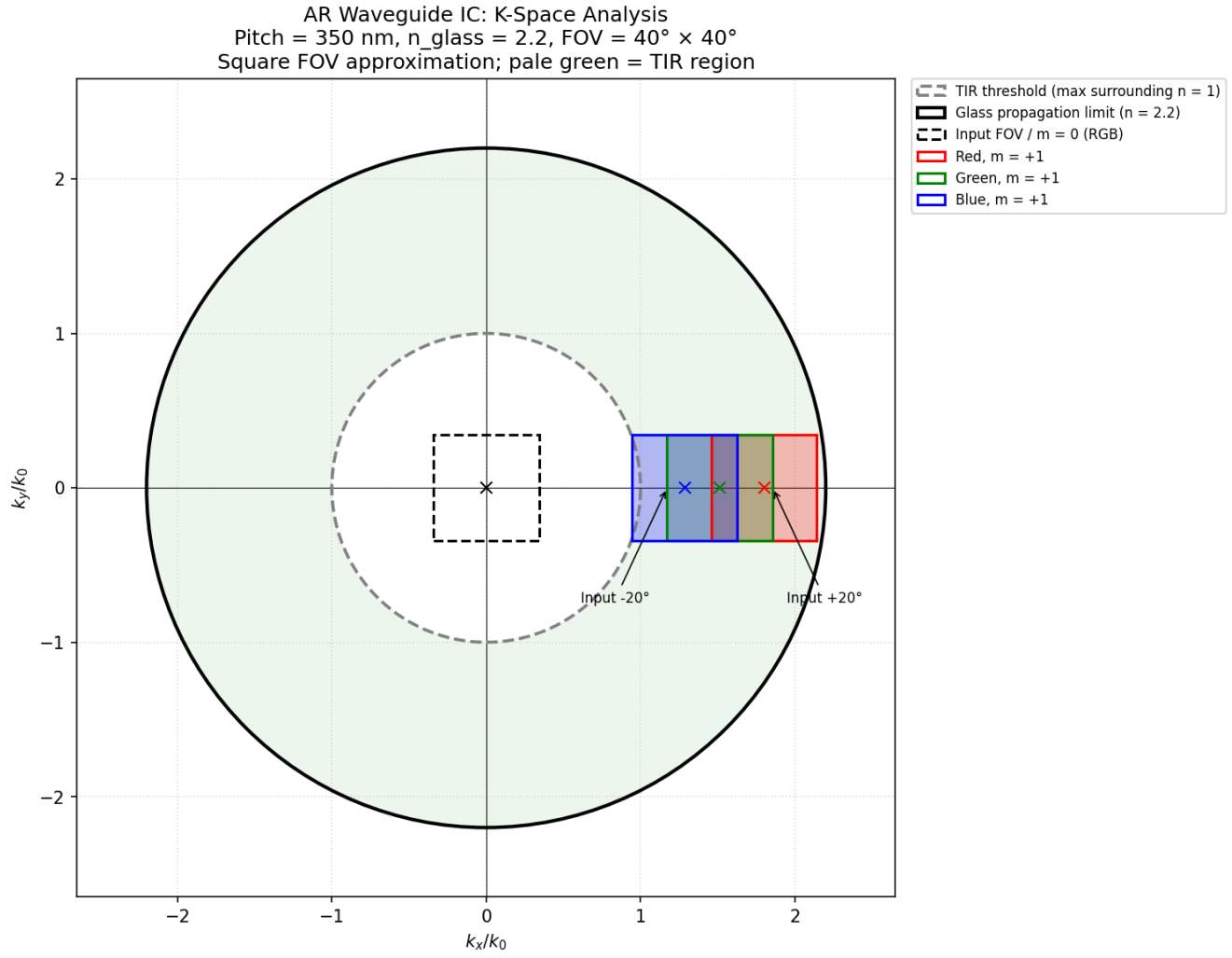


Start with blank fields. Select the incident side and diffraction orders manually.

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5. Read the diagram



Pale green annulus: TIR region.

Black dashed square: input FOV / m = 0.

RGB squares: shifted diffraction orders.

Shift in u = mλ / pitch. The square size stays fixed.

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6. TIR region and radial distance

$$\rho = \sqrt{\left(\frac{k_x}{k_0}\right)^2 + \left(\frac{k_y}{k_0}\right)^2} = \frac{\sqrt{k_x^2 + k_y^2}}{k_0}, \qquad k_0 = \frac{2\pi}{\lambda_0}$$

max(n_top, n_bottom) < ρ < n_glass

ρ is dimensionless. λ_0 is the vacuum wavelength.

Quantity	Interpretation
ρ_min	Nearest point of the FOV square to the origin
ρ_max	Farthest point of the FOV square from the origin
Entire FOV passes	ρ_min exceeds the inner limit AND ρ_max is below the outer limit

The boundary inequalities are strict. Touching either circle does not count as full-FOV TIR.

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7. Example classification

Color / order	ρ_min	ρ_max	Result
RGB, m = 0	0.00000	0.48369	No strict TIR at both interfaces
Red, m = +1	1.45798	2.16915	Entire FOV within TIR
Green, m = +1	1.17227	1.88755	Entire FOV within TIR
Blue, m = +1	0.94369	1.66328	Crosses inner boundary

TIR region: 1 < ρ < 2.2

Example: Glass n = 2.2, FOV = 40°, pitch = 350 nm, RGB = 630 / 530 / 450 nm.

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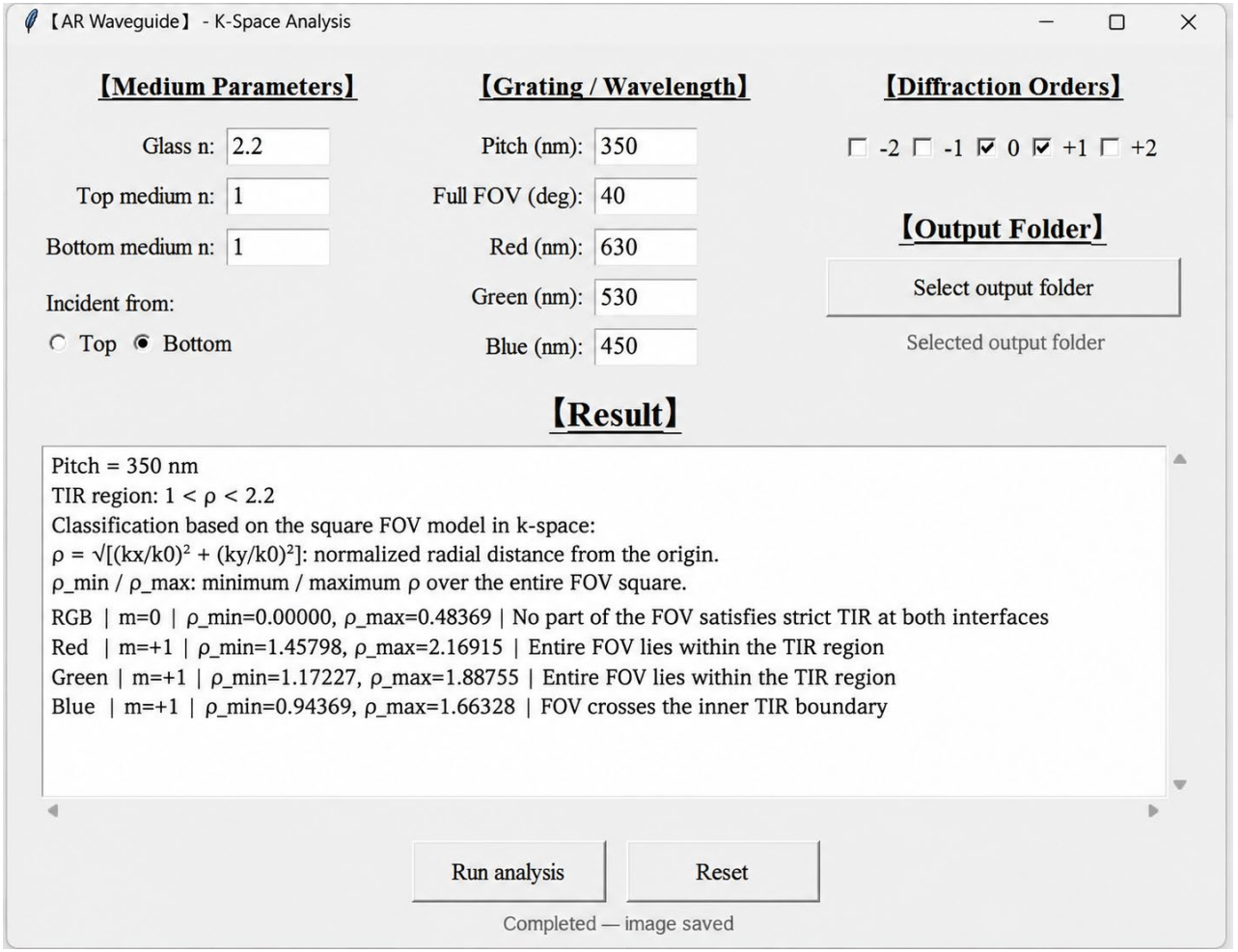
8. Saved image and result text

One 360 dpi JPG is saved directly in the selected folder.

Filename:
K_space_IC_all_orders_<timestamp>.jpg

Result text shows the FOV size, TIR limits and per-order classification.

The plot opens in a separate preview window.



Illustrative result view with the current Windows layout

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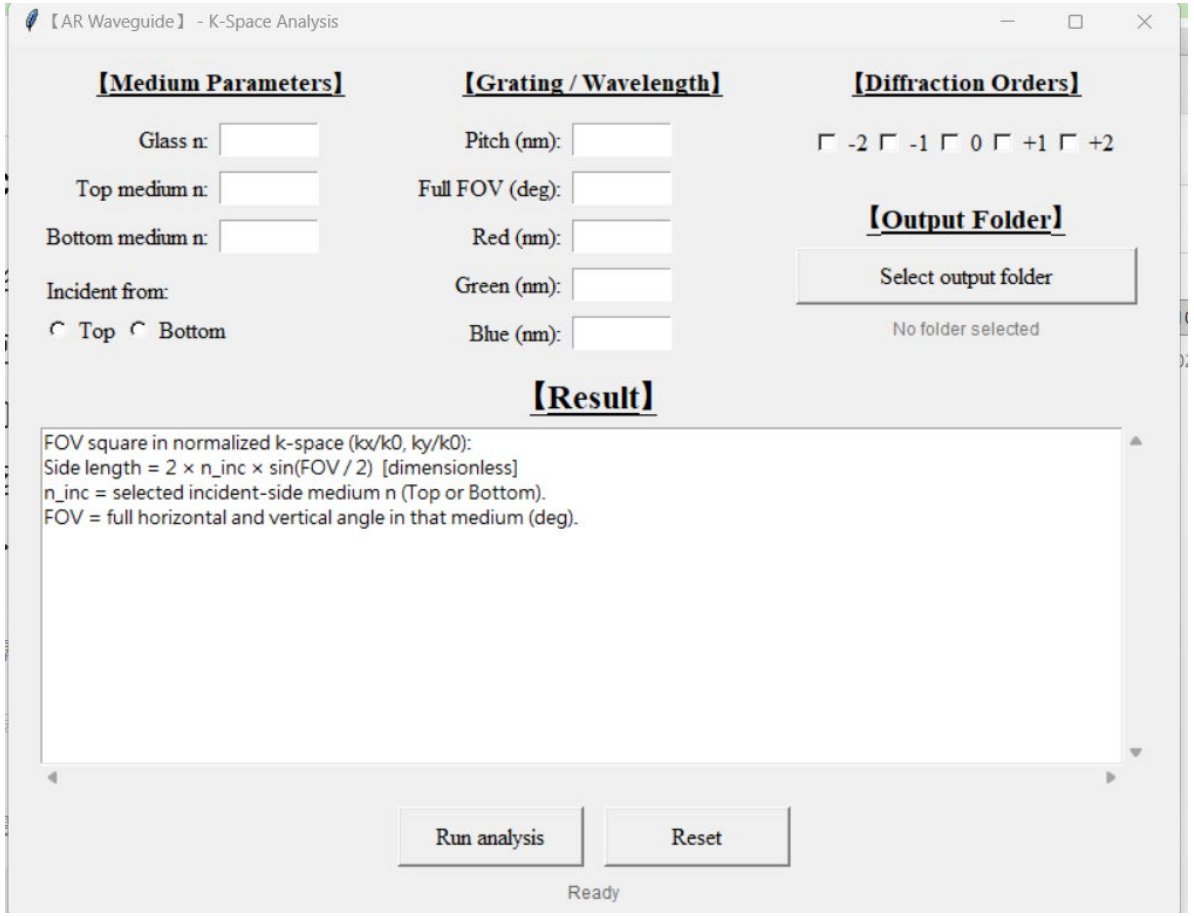
9. Reset and input checks

Reset clears numeric inputs, the incident-side choice and all order selections.

Reset clears the selected folder and closes the plot preview. Saved files remain.

Use positive finite indices, wavelengths and pitch.

For this square-FOV model, use 0° < FOV < 90°. Larger values require a different mapping.



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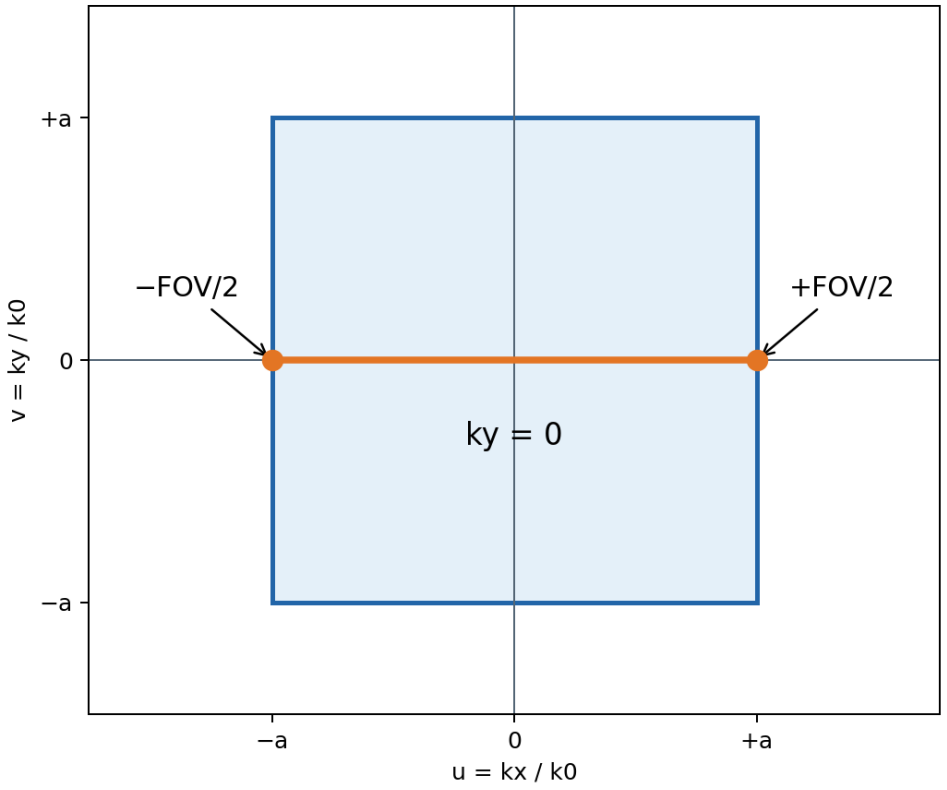
10. Scope of the analysis

The model assumes a square FOV centered at normal incidence.

The grating changes kx; ky remains unchanged.

A single Glass n is used for all three wavelengths.

TIR feasibility does not give diffraction efficiency or transmitted power.



Use this GUI to check whole-square direction coverage before examining pitch sensitivity.

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Quick reference

Step	Action
Input	Fill Glass / Top / Bottom n, pitch, FOV and RGB wavelengths
Select	Choose the incident side and at least one diffraction order
Run	Select output folder, then click Run analysis
Review	Inspect the RGB squares, ρ_min / ρ_max and TIR classification
Save / reset	JPG saves automatically; Reset prepares the next case

Launch: python K_space_GUI.py | Dependencies: numpy, matplotlib and Tkinter

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